Module8_lab

October 18, 2026

[1]:
```
# Step 1 - Open Colab and load the penguins
```

[2]:
```
import pandas as pd
```

[3]:
```
df=pd.read_csv("Downloads/penguins.csv")
```

[4]:
```
# Step 2 - Establish ground truth by hand, no AI
```

[5]:
```
df.shape
```

[5]:
```
(344, 8)
```

[6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    str
 1   island             344 non-null    str
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    str
 7   year               344 non-null    int64
dtypes: float64(4), int64(1), str(3)
memory usage: 27.6 KB
```

[7]:
```
df.describe()
```

[7]:

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	\
count	342.000000	342.000000	342.000000	342.000000	
mean	43.921930	17.151170	200.915205	4201.754386	
std	5.459584	1.974793	14.061714	801.954536	
min	32.100000	13.100000	172.000000	2700.000000	
25%	39.225000	15.600000	190.000000	3550.000000	
50%	44.450000	17.300000	197.000000	4050.000000	

```
75%         48.500000      18.700000         213.000000  4750.000000
max         59.600000      21.500000         231.000000  6300.000000

              year
count   344.000000
mean   2008.029070
std       0.818356
min    2007.000000
25%    2007.000000
50%    2008.000000
75%    2009.000000
max    2009.000000
```

[8]:
```
print(df.isnull().sum())
```

```
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64
```

[9]:
```
print(df["species"].value_counts())
```

```
species
Adelie       152
Gentoo       124
Chinstrap     68
Name: count, dtype: int64
```

[10]:
```
print(df.groupby("species")["flipper_length_mm"].mean())
```

```
species
Adelie       189.953642
Chinstrap    195.823529
Gentoo       217.186992
Name: flipper_length_mm, dtype: float64
```

[11]:
```
# Step 3 - Prompt the AI (and log it)
```

Prompt Log

You are helping with exploratory data analysis. My CSV has columns: species (object), island (object), bill_length_mm (float64), bill_depth_mm (float64), flipper_length_mm (float64), body_mass_g (float64), sex (object), year (int64). Write Python using pandas and seaborn to: * (1) summarize missing data, * (2) compare flipper_length_mm across species, * (3) produce one labeled chart.

Explain each step in comments.

[12]:
```
# Step 4 - Run the AI's code
```

[13]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

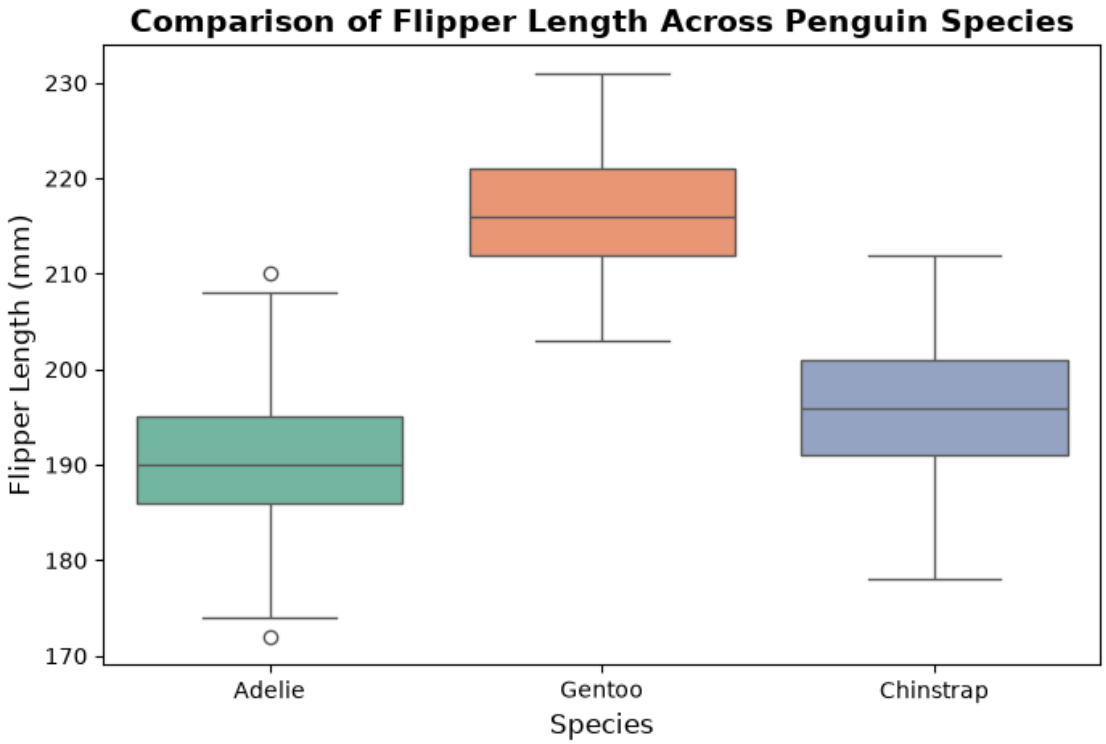
[14]:
```
# (1) Summarize missing data
print("Missing Data Summary:")
print(df.isnull().sum())

# (2) Compare flipper_length_mm across species
species_flipper_summary = df.groupby("species")["flipper_length_mm"].agg(
    ["count", "mean", "median", "std"]
)
print("\nFlipper Length Comparison by Species:")
display(species_flipper_summary)

# (3) Produce one labeled chart
plt.figure(figsize=(8, 5))
sns.boxplot(
    data=df,
    x="species",
    y="flipper_length_mm",
    hue="species",
    palette="Set2",
    legend=False,
)
plt.title(
    "Comparison of Flipper Length Across Penguin Species",
    fontsize=14,
    fontweight="bold",
)
plt.xlabel("Species", fontsize=12)
plt.ylabel("Flipper Length (mm)", fontsize=12)
plt.show()
```

```
Missing Data Summary:
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64
```

```
Flipper Length Comparison by Species:
           count        mean  median       std
species
Adelie       151  189.953642   190.0  6.539457
Chinstrap     68  195.823529   196.0  7.131894
Gentoo       123  217.186992   216.0  6.484976
```



[15]:
```
# Step 5 -  verification check
```

[16]:
```
ai_comparison_check = df.groupby("species")["flipper_length_mm"].mean()
print("Verification Check - Mean Flipper Length:")
print(ai_comparison_check)
```

```
Verification Check - Mean Flipper Length:
species
Adelie       189.953642
Chinstrap    195.823529
Gentoo       217.186992
Name: flipper_length_mm, dtype: float64
```

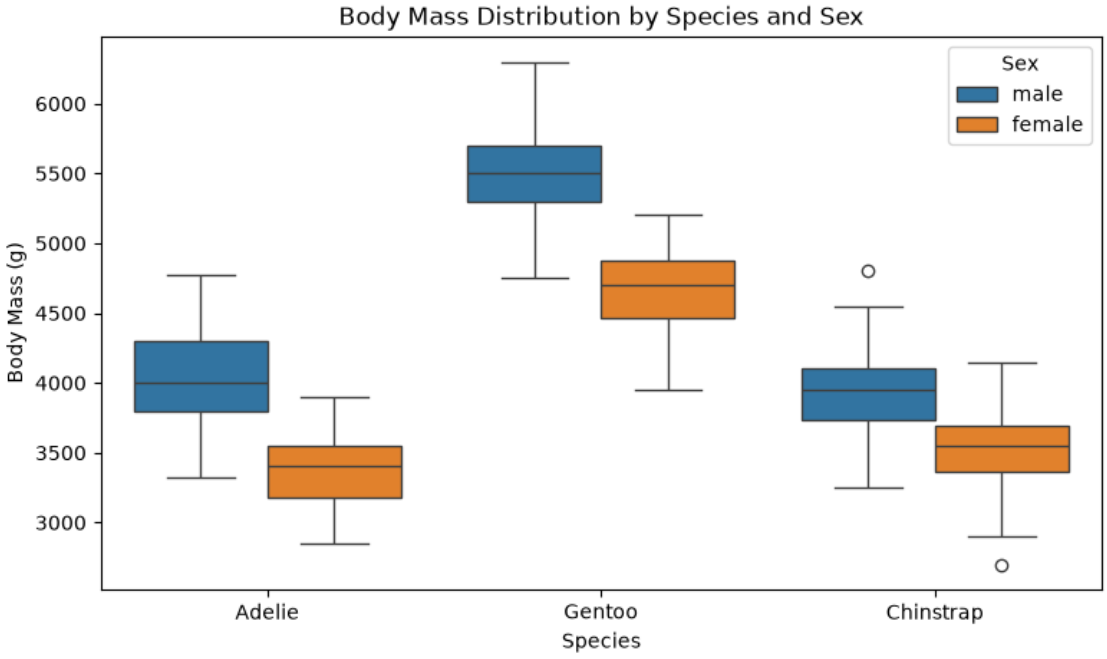
My hand-computed groupby mean flipper lengths (Adelie: 189.95, Chinstrap: 195.82, Gentoo: 217.19) perfectly match the values produced by the AI's script. It is also verified by the df.describe section when I was checking the data. Both methods look to drop the 2 missing rows without causing discrepancies.

[17]:
```
# Step 6: One follow-up, one judgment
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[18]:
```
# Follow-up - looking at body mass across the species and sex with a box plot
```

[19]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(9, 5))
sns.boxplot(
    data=df, x="species", y="body_mass_g", hue="sex")
plt.title("Body Mass Distribution by Species and Sex")
plt.xlabel("Species")
plt.ylabel("Body Mass (g)")
plt.legend(title="Sex")
plt.show()
```



[20]:
```
# Step 7: Your original insight, no AI
```

Across all species, the male penguins are heavier than females, showing a clear weight gap between the sexes. Additionally, Gentoo penguins are much heavier than both Adelie and Chinstraps penguins that even female Gentoos weigh more than the males of the other two species.

[21]:
```
df.groupby(["species", "sex"])["body_mass_g"].mean()
```

[21]:
```
species    sex
Adelie     female    3368.835616
```

```
            male      4043.493151
Chinstrap   female    3527.205882
            male      3938.970588
Gentoo      female    4679.741379
            male      5484.836066
Name: body_mass_g, dtype: float64
```

[22]:
```
# Step 8: Privacy reflection (markdown cell)
```

One dataset from my own work or life that I must never paste into a public AI tool is a database containing sensitive employee information and records. Doing so would violate company I.T. policies and privacy laws. Instead, I would try to group the data by department brackets, or process that would make it more generalized without the senitive information to be used for AI analysis.

[ ]:
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